

Riemann's Definitive Integral Definition

Let $f : [a, b] \rightarrow \mathbb{R}$ be a bounded function. We say that f is integrable over $[a, b]$ if $\exists I(f) \in \mathbb{R}$ with the given property :

For every $\epsilon > 0$ we can find $\delta > 0$ such that : if $P = \{a = x_0 < x_1 < \dots < x_n = b\}$ is a partition of $[a, b]$ with $\|P\| < \delta$ and if $\xi_k \in [x_k, x_{k+1}]$, where $k = 0, 1, 2, \dots, n-1$ is a random choice of points from the subintervals that P defines, then :

$$\left| \sum_{k=0}^{n-1} f(\xi_k)(x_{k+1} - x_k) - I(f) \right| < \epsilon$$

In this case we say that $I(f)$ is the (R) -integral of f in $[a, b]$.

It is important to define Ξ which is the choice of random points $\{\xi_0, \xi_1, \dots, \xi_k\}$ as $\sum(f, P, \Xi)$ is defined :

$$\sum_{k=0}^{n-1} f(\xi_k)(x_{k+1} - x_k)$$

Observe how Ξ “intrudes into” $\sum(f, P, \Xi)$ because for the same partition P we can have many different options $\Xi = \{\xi_0, \xi_1, \dots, \xi_k\}$ with $\xi \in [\xi_k, \xi_{k+1}]$.

This way we uncover the core idea behind the definition which is :

$$\int_a^b f(x) dx = \lim_{\|P\| \rightarrow 0} \sum(f, P, \Xi)$$